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Enhancing fine-grained image classification through attentive batch training

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ABSTRACT

Fine-grained image classification, which is a challenging task in computer vision, requires precise differentiation among visually similar object categories. In this paper, we design a novel framework, namely Relationship Batch Integration (RBI), allowing the discernment of vital visual features that may remain elusive when examining a singular image representative of a particular class. Our proposed method, validated through extensive experiments, significantly boosts the accuracy of fine-grained classifiers, achieving state-of-the-art performance with (97.79%) on the Stanford Dog dataset, even attaining a top result of (93.71%) on the Tiny-Imagenet dataset for general image classification.

1. Introduction

Fine-grained classification is an important task in computer vision as it has a wide range of real-world applications, including image recognition, disease diagnosis [1,2], or biodiversity monitoring [3,4], where distinguishing between visually similar subcategories is crucial. With the rapid advancement of technology, we now have the capability to collect and store a large amount of image data from various sources. Nevertheless, the classification of objects in images characterized by a high degree of similarity, commonly referred to as the *fine-grained image classification* problem — examples include the categorization of bird species, types of leaves, or models of electronic products — presents a noteworthy and persistent challenge. In addition to image classification in general, fine-grained image classification exposes more significantly challenging, including (i) substantial intra-class variation, with objects in the same category exhibiting significant pose and viewpoint differences; (ii) subtle inter-class distinctions, where objects from different categories may closely resemble each other with minor differences; (iii) constraints on training data, as labeling fine-grained categories often demands specialized expertise and a substantial amount of annotation effort [5]. For these reasons, fine-grained classification remains an open topic for research community.

Motivated by the fact that deep neural networks (DNNs) encounter challenges in effectively distinguishing intricate features and grappling with the inherent complexities of learning detailed patterns, our study centers on scrutinizing the mechanisms by which image feature extractors discern nuanced features. Additionally, our investigation delves into the process of *explicitly* amalgamating these discerned features across a batch of images to construct intricate feature maps, ultimately enhancing the accuracy of fine-grained image classification. Fig. 1 illustrates an example of our idea, integrating subtle features extracted from several images to generate a sophisticated feature. In the figure, the delineated bounding boxes within each image delineate regions of interest identified by the model for focused attention. Boxes sharing the same color signify images belonging to identical classes, while distinct colors indicate diverse classification categories. We aim at discerning specific connections between the reference image and others, facilitating the amalgamation of subtle features derived from various images, even those belonging to different categories, to complement the features of the target image. Concretely, as exemplified in the provided figure, features corresponding to the snout, ear, and eye regions extracted from three distinct images are harmonized with the feet region. This integration results in a cohesive feature representation that concentrates on multiple salient positions of the dog within the target image. Intuitively, we can design a novel framework for this purpose, termed as Relationship Batch Integration (RBI).

Our contributions can be summarized in the following:

- We present a novel attention module, termed Residual Relationship Attention (RRA), that effectively exploits subtle features from multiple images and combines them to generate sophisticated features, particularly in the case of a CNN-based backbone. Additionally, it improves the focusing ability on important features when utilizing an Attention-based backbone. To this end,

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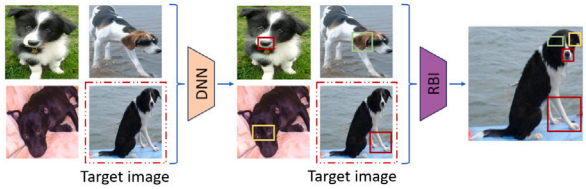


Fig. 1. Example of intra-batch feature fusion to enhance predictivity for target images.

it addresses the issue of ambiguity in fine-grained classification, mitigating its impact.

- We introduce a novel technique called Relationship Position Encoding (RPE), which effectively preserves the relationship information between images within a batch.
- We introduce a novel RBI framework that integrates RRA and RPE, offering seamless integration into fine-grained classifiers. Remarkably, even when employing a reduced version of the pre-trained backbone with fewer parameters compared to the larger baselines, the RBI improves both accuracy and processing time.
- Our extensive experiments on publicly available datasets demonstrate the model's capability to enhance feature clustering and accuracy, while also achieving state-of-the-art results on the CUB200-2011 and Stanford Dogs datasets.

2. Proposed approach

2.1. Relationship Batch Integration (RBI) Framework

As illustrated in Fig. 2, the Relationship Batch Integration (RBI) Framework is composed of three components: *relationship position encoding (RPE)*, *deep neural network (DNN) feature extractor*, and *residual relationship attention (RRA) module*. Specifically, for a given batch of input images $\mathbf{X} \in \mathbb{R}^{B\times 3\times H\times W}$ (B denote the number of images within the batch, while H and W represent the height and width of an image in $\mathbf{X}$, respectively), we begin by calculating the similarity between each pair of input images using the relationship position encoding module, described in details in Section 2.2. This process produces a similarity matrix $\mathbf{S} \in \mathbb{R}^{B\times B}$. Simultaneously, $\mathbf{X}$ is passed through the DNN feature extractor, which can be a CNN-based backbone such as Densenet [6], ConvNeXt [7], or an Attention-based backbone such as Swin Transformer [8], ViT [9]. This yields DNN feature embeddings $\mathbf{N} \in \mathbb{R}^{B\times D}$ (where D signifies the dimensionality of the DNN feature embeddings). Subsequently, the similarity matrix $\mathbf{S}$ and DNN feature embeddings $\mathbf{N}$ are utilized as inputs for the RRA module. Finally, the output of the RRA module is forwarded to fully connected layers to generate logits for classification.

2.2. Relationship Position Encoding (RPE)

Contrary to prior studies such as [10], which commonly utilize position encoding to incorporate information about the relative or absolute position of tokens in a sequence, the order of images in the batch is not essential in this case. This is attributed to the random sampling of images from the dataset, devoid of any inherent relational structure. However, it is important to preserve the position of relationships between images within the batch during the integration process within the RRA module. Thus, we devise a Relationship Position Encoding to serve the crucial purpose of embedding this positional insight into the RRA module.

In this study, the similarity weight s_{ij} ($s_{ij} \in \mathbf{S}$) between two images is determined by RPE. To begin, we compute the Mean Squared Error (MSE) between two images $\mathbf{I}_i$ and $\mathbf{I}_j$ within a batch (where $\mathbf{I}_i, \mathbf{I}_j \in \mathbb{R}^{3\times H\times W}$):

$$\mathrm{MSE}(\mathbf{I}_i, \mathbf{I}_j) = \frac{1}{3HW} \sum_{x=0}^{2} \sum_{y=0}^{H-1} \sum_{z=0}^{W-1} \left[\mathbf{I}_i(x,y,z) - \mathbf{I}_j(x,y,z)\right]^2. \tag{1}$$

Then, we compute s_{ij} in the form of a normalized PSNR by incorporating a very small coefficient ϵ to prevent division by zero when the two images are identical:

$$s_{ij} = \widehat{\mathrm{PSNR}}(\mathbf{I}_i, \mathbf{I}_j) = 20 \cdot \log_{10}\left(\frac{\mathrm{MAX}_{\mathbf{I}_i}}{\sqrt{\mathrm{MSE}(\mathbf{I}_i, \mathbf{I}_j)} + \epsilon}\right); \tag{2}$$

where $\mathrm{MAX}_{\mathbf{I}_i}$ represents the maximum possible pixel value of $\mathbf{I}_i$.

2.3. Residual Relationship Attention (RRA)

Before delving deeper into the RRA structure, we first establish several definitions for mathematical operations.

Vertical Duplication. Given DNN feature embeddings $\mathbf{N} \in \mathbb{R}^{B\times D}$ as input. Initially, the operation involves expanding the tensor along its first dimension, transforming it from $\mathbf{N} \in \mathbb{R}^{B\times D}$ to $\mathbf{N} \in \mathbb{R}^{1\times B\times D}$. Subsequently, the vertical duplication operation duplicates each instance along the first dimension. It concatenates them along the same dimension, resulting in a tensor of shape $\mathbb{R}^{B\times B\times D}$, formulated as:

$$\mathcal{D}_v(\mathbf{N}) = \|_{i=1}^{B} \mathbf{N} \in \mathbb{R}^{1\times B\times D} \rightarrow \mathbb{R}^{B\times B\times D}. \tag{3}$$

Horizontal Duplication. The horizontal duplication operation involves unsqueezing the second dimension of the input and concatenating the duplicated inputs along the same dimension:

$$\mathcal{D}_h(\mathbf{N}) = \|_{i=1}^{B} \mathbf{N} \in \mathbb{R}^{B\times 1\times D} \rightarrow \mathbb{R}^{B\times B\times D}. \tag{4}$$

Depth Sum. The operation of depth sum involves the reduction of the input matrix $\mathbf{F}$, which is of dimension $B\times B\times D$, through summation along its last dimension. Mathematically, it can be expressed as:

$$\mathcal{S}_d(\mathbf{F}) = \left[\sum_{k=1}^{D} f_{ijk}\right]_{i,j=\overline{1,B}} \in \mathbb{R}^{B\times B}; \tag{5}$$

where f_{ijk} represents the value of the element at position i, j, k in the matrix $\mathbf{F}$.

Vertical Sum. Similarly, the vertical sum operation entails reducing the first dimension of the input matrix $\mathbf{F}$ through summation. It is represented as:

$$\mathcal{S}_v(\mathbf{F}) = \left[\sum_{i=1}^{B} f_{ijk}\right]_{j=\overline{1,B},k=\overline{1,D}} \in \mathbb{R}^{B\times D}. \tag{6}$$

Next, we proceed to explore the individual components of the RRA structure in greater detail.

Keys. The keys in the RRA are derived from the parameter matrix of keys, denoted as $\mathbf{W}_K \in \mathbb{R}^{D\times D}$, and DNN feature embeddings $\mathbf{N} \in \mathbb{R}^{B\times D}$:

$$\mathbf{K} = \mathcal{D}_v(\mathbf{N}\mathbf{W}_K) \in \mathbb{R}^{B\times B\times D}. \tag{7}$$

Queries, Values. Given weight matrices $\mathbf{W}_Q \in \mathbb{R}^{D\times D}$ and $\mathbf{W}_V \in \mathbb{R}^{D\times D}$ representing learnable parameters of queries and values, respectively, and DNN feature embeddings $\mathbf{N} \in \mathbb{R}^{B\times D}$, the queries $\mathbf{Q}$ and the values $\mathbf{V}$ in the RRA are calculated according to the following equations:

$$\mathbf{Q} = \mathcal{D}_h(\mathbf{N}\mathbf{W}_Q) \in \mathbb{R}^{B\times B\times D}; \tag{8}$$

$$\mathbf{V} = \mathcal{D}_h(\mathbf{N}\mathbf{W}_V) \in \mathbb{R}^{B\times B\times D}. \tag{9}$$

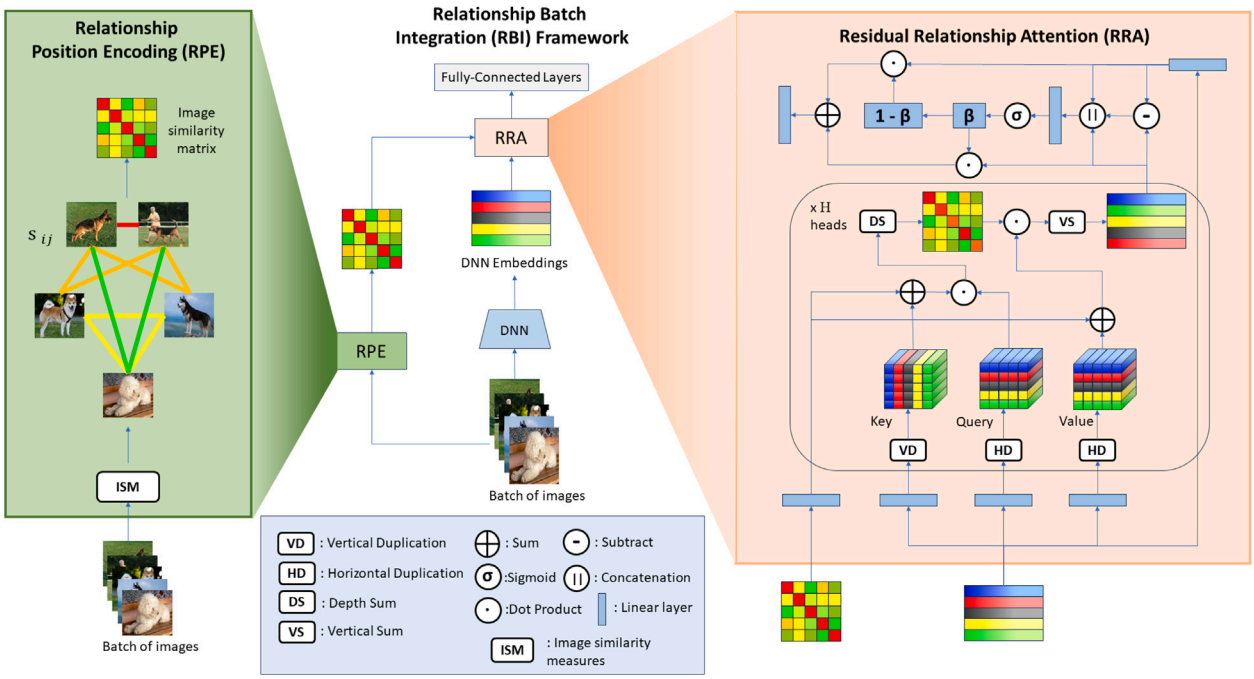


Fig. 2. Relationship Batch Integration (RBI) Framework.

Attention. To compute the attention matrix $\mathbf{A}$, given the relationship position embeddings $\mathbf{S} \in \mathbb{R}^{B\times B}$, queries $\mathbf{Q}$, and keys $\mathbf{K}$, the following equation is utilized:

$$\mathbf{A} = \mathrm{softmax}\left(\mathcal{S}_d\left(\frac{\mathbf{Q}\cdot(\mathbf{K}+\mathbf{S})}{\sqrt{D}}\right)\right) \in \mathbb{R}^{B\times B}. \tag{10}$$

Attention Embeddings. The attention embeddings $\mathbf{Z}$ are computed using the attention matrix $\mathbf{A}$, values $\mathbf{V}$, and relationship position embeddings $\mathbf{S}$:

$$\mathbf{Z} = \mathcal{S}_v\big(\mathbf{A}\cdot(\mathbf{V}+\mathbf{S})\big) \in \mathbb{R}^{B\times D}. \tag{11}$$

Output. Finally, the output $\mathbf{C}$ of the RRA is computed using attention embeddings $\mathbf{Z}$, parameter matrix of residual $\mathbf{W}_S \in \mathbb{R}^{D\times D}$, and DNN feature embeddings $\mathbf{N}$:

$$\beta = \mathrm{sigmoid}\Big(\mathbf{W}_\beta\big[\mathbf{Z}||(\mathbf{W}_S\mathbf{N})||(\mathbf{Z}-\mathbf{W}_S\mathbf{N})\big]\Big); \tag{12}$$

$$\mathbf{C} = \mathrm{BatchNorm}\big((1-\beta)\mathbf{Z}+\beta\mathbf{W}_S\mathbf{N}\big); \tag{13}$$

where || represents the concatenation operation along the last dimension of tensors and $\mathbf{W}_\beta$ denotes the parameter matrix of beta. Eqs. (12) and (13) facilitate the preservation of the original features of the DNN and enable flexible adjustment of the weights assigned to both the DNN and RRA features.

3. Experiments

3.1. Comparison to existing methods

Baselines. We perform experiments on three well-known fine-grained datasets: CUB-200-201 [11], Stanford Dogs [12], and NABirds [13].[1] To validate the effectiveness and generalization of our method, we investigate the performance of incorporating RBI on four different well-known DNNs and their variants, including DenseNet [6], MobileNet [14], ConvNeXt [7], SwinTransformer [8], and HERB [15]. It is important to highlight that our RBI is the only modification, while all other training configurations and hyperparameters remain unaltered from the original implementations. Even though we incorporate our proposed method across various techniques and assess it on diverse datasets, we maintain the consistent parameter configuration detailed earlier throughout all experiments.

Comparison results. Table 1 shows the impact of our RBI on fine-grained classification performance across different methods and datasets. Our interesting findings are summarized as follows:

- The table clearly illustrates that the incorporation of RBI consistently improves fine-grained classification results. Notably, we observe an average increase of +2.78%, +3.83%, and +3.29% on the Stanford Dogs, CUB-200-2011 datasets and NABirds, respectively.[2]
- Improving existing fine-grained classification methods is a challenging endeavor. However, as shown in Table 1, our proposed approach achieves new state-of-the-art results on the Stanford Dogs dataset.[3] It is crucial to note that the reported accuracies on the CUB-200-2011 and NABirds datasets diverge from those in the literature, such as HERB, TransFG, and P2P-Net, due to our uniform training configurations outlined in Section 3, which may differ from the original methodologies.

[1] For a more comprehensive understanding of the detailed information about the datasets and implementation details, please refer to Section Datasets and Experimental Settings in Supplementary Material.

[2] Changing the batch size during the testing process does not affect the results. For more details, please refer to Section The Impact of Batch Configurations in the Supplementary Material.

[3] According to the comparison table in https://paperswithcode.com/sota/fine-grained-image-classification-on-stanford-1 on 01/02/2025.

Table 1
The impact of RBI on fine-grained classification outcomes when incorporated into various DNN techniques. The accuracy gain when applying RBI is provided in the brackets. The bolded models highlight how the RBI model with a smaller pretrained backbone outperforms the baseline large model in both inference time and parameter count.

Method	Inference time	# params	Acc (%)		
			Stanford Dogs	CUB-200-2011	NABirds
MobilenetV3-S	0.013	1.6M	73.12	67.5	66.46
MobilenetV3-S-RBI	0.016	17.4M	77.01(+3.89)	69.86(+2.36)	69.1(+2.64)
MobilenetV3-L	0.035	4.4M	78.31	77.65	75.86
MobilenetV3-L-RBI	0.039	23.2M	82.72(+4.41)	80.77(+3.12)	79.82(+3.96)
Densenet201	0.28	18.3M	83.95	79.13	77.55
Densenet201-RBI	0.29	73.7M	87.72(+3.77)	84.48(+5.35)	83.81(+6.26)
Densenet161	0.42	26.7M	84.46	79.68	78.97
Densenet161-RBI	0.45	88.7M	88.47(+4.01)	84.79(+5.11)	84.75(+5.78)
SwinT-Small	0.51	49.1M	91.39	86.27	86.74
SwinT-Small-RBI	0.52	61.7M	92.79(+1.40)	87.35(+1.08)	87.97(+1.23)
SwinT-Big	0.82	87.0M	92.11	85.86	86.32
SwinT-Big-RBI	0.84	102.8M	93.06(+0.95)	87.9(+2.04)	88.03(+1.71)
ConvNeXtBase	0.59	88.7M	94.77	81.93	85.31
ConvNeXtBase-RBI	0.61	103.4M	96.56(+1.79)	87.52(+5.59)	87.86(+2.55)
ConvNeXtLarge	1.22	197.9M	95.71	81.74	85.53
ConvNeXtLarge-RBI	1.23	231.8M	97.79(+2.08)	87.8(+6.06)	88.11(+2.58)
HERB-SwinT	1.74	286.6M	88.62	89.9	90
HERB-SwinT-RBI	1.88	318.2M	88.9(+0.28)	90.37(+0.47)	90.61(+0.61)
TransFG	0.85	86.3M	89.18	90.19	89.9
TransFG-RBI	0.86	95.2M	89.76(+0.58)	90.66(+0.47)	90.45(+0.55)
P2P-Net	0.46	63.4M	83.14	86.07	85.12
P2P-Net-RBI	0.48	110.6M	89.3(+6.16)	91.07(+5.00)	90.56(+5.44)
Avg. Improvement			**+2.51**	**+3.46**	**+3.04**

- Additionally, we observe that for some models, when we add the RBI module to smaller variants, they achieve better accuracy than the larger variants without the module, while also being less time-consuming and complex. For instance, SwinT-Small-RBI (61.7M parameters) outperforms SwinT-Big (87M parameters), and ConvNeXtBase-RBI (103.4M parameters) surpasses ConvNeXtLarge (197.9M). This partly demonstrates the effectiveness of the proposed module when integrated into different backbones. Regarding neural network complexity, despite a significant increase in the number of parameters in the proposed models compared to the base ones, the inference time varies only slightly between them.
- Furthermore, our observations reveal that the inclusion of RBI in all smaller model versions outperforms their larger counterparts in terms of inference time. This comparison is fairer than solely considering the number of parameters, especially for compact models like MobileNet and DenseNet, which exhibit a remarkable speed improvement of up to twice as fast.

In summary, our proposed approach consistently demonstrates enhanced performance across various classifiers and fine-grained datasets. Moreover, our method can easily integrate with cutting-edge classifiers to yield further enhancements. Notably, the parameter configuration for our approach remains uncomplicated, delivering favorable outcomes with a single setup across diverse classifiers and datasets.

3.2. Feature extracted by conventional DNN and RBI

Recall that, in our proposed RBI framework, we train the DNN and RRA components simultaneously within an end-to-end framework, enabling the direct influence of RRA backpropagation on DNN parameters during training. As a result, features generated by a conventional DNN backbone exhibit distinct characteristics compared to those produced by the DNN component in the RBI method. Specifically, the RRA tends to propagate gradients backward, allowing the DNN to discern subtler features for combinatorial purposes and create intricate, rich feature maps. Conversely, traditional DNN parameters are propagated to emphasize the most critical feature, assisting in the classification of images into distinct classes.

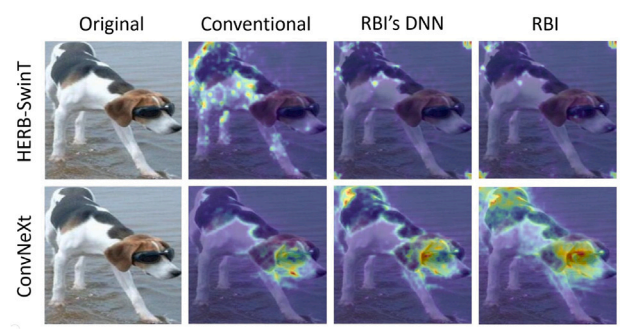
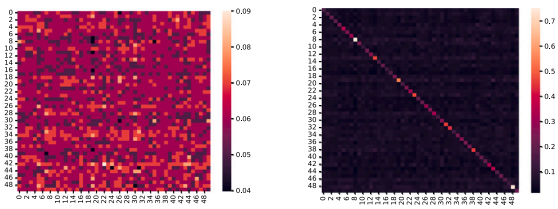


Fig. 3. Comparison between features extracted by ConvNeXt-Large, ConvNeXt-Large-RBI, HERB-SwinT and HERB-SwinT-RBI on Stanford Dogs dataset, illustrated by GradCam.

GradCAM [16] visualizations in Fig. 3 highlight aspects of the aforementioned argument. Here, we employ ConvNeXt-Large, and HERB-SwinT as the DNN backbone. The top row presents GradCAM visualizations for HERB-SwinT, while the bottom row illustrates ConvNeXt visualizations. Moving from left to right in each row, the columns depict features from a traditional method, features extracted by RBI's DNN, and the combined features of both DNN and RRA.

In ConvNeXt, the traditional method predominantly highlights the dog's head in its feature map, showing less emphasis on colored regions elsewhere on the dog's body. Conversely, employing the RBI method's DNN results in a feature map that captures subtle features across various positions, including the head, colored regions, and even the hips. When examining Attention-based models like HERB-SwinT, the RBI method diverges from conventional approaches by focusing on



(a) Similarity matrix without RPE (b) Similarity matrix with RPE

Fig. 4. The heatmap of the similarity matrix representing the relationship between images in a batch generated by RRA.

eliminating redundant features and concentrating solely on specific, subtle features crucial for fine-grained image classification tasks.

3.3. RRA similarity matrix

In this section, we explore two scenarios: RRA with and without RPE. More specifically, the computation of the similarity weight s_{ij} between two images. The experimental result is visualized in Fig. 4, where the x-axis and y-axis enumerate the image indices within a batch, and each cell corresponding to positions on the x and y axes signifies the connection between the two images. The color intensity mirrors the strength of relationships between images, with brighter hues denoting stronger connections and darker shades indicating weaker ones. The matrix observed from Fig. 4(a) that, without RPE, lacks intuitiveness, whereas the matrix generated by RRA with RPE in Fig. 4(b) is more easily comprehensible. This is evident from the prominent values along the diagonal and the symmetrical appearance of the matrix across the main diagonal. There are two benefits to having diagonally symmetrical elements and larger diagonal elements in the similarity matrix. Firstly, it enhances the understanding and intuitiveness of how RRA synthesizes images in batches. Secondly, it leads to a reduced loss of original image information, resulting in a slight improvement in accuracy.[4]

4. Conclusion

In this study, we has proposed a novel framework that, while deceptively straightforward, has not been explored in prior works. Empirical experiments on benchmark datasets are conducted to show the effectiveness and to demonstrate seamless integration with various fine-grained classifiers for significant accuracy improvements of our proposed approach. Our architectural innovation also significantly reduces model parameters and inference latency compared to conventional DNN architectures.

CRediT authorship contribution statement

Duy M. Le: Writing – review & editing, Writing – original draft, Methodology, Conceptualization, Data curation, Formal analysis, Investigation, Visualization. **Bao Q. Bui:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Resources, Software. **Anh Tran:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Formal analysis, Data curation. **Cong Tran:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Formal analysis, Funding acquisition. **Cuong Pham:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Formal analysis, Funding acquisition.

[4] Despite the quadratic increase in the size of the similarity matrix with batch size, which could potentially lead to memory constraints when dealing with very large batches, this issue is not a significant bottleneck in practical applications. A more detailed discussion on this matter is provided in the Memory Usage section of the supplementary material.

Declaration of competing interest

The authors declare that there is no conflict of interest in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.icte.2025.10.002.

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